

Modelling this horn in HFSS

Optimum pyramidal horn, WR-340 feed, 2.45 GHz · every dimension generated from antenna/horn.py

A procedure, not a tutorial on the software. It builds the exact antenna this repo specifies, in the order that gets you a trustworthy S_{11} and pattern with the fewest re-solves, and it ends by checking the solver against closed-form theory — because a converged answer and a correct answer are different things.

solver	Ansys HFSS (Electronics Desktop). Menu paths are 2023 R2; older versions differ slightly
design	optimum pyramidal horn, 13.4 dBi, WR-340 feed
feed	coax probe, $\lambda_g/4$ backshort, SMA
source	antenna/horn.py — re-run make_hfss.py if that changes

0 · The numbers to beat

Write these down before you open HFSS. They come from closed-form optimum-horn theory, they are independent of the solver, and at the end you will compare the simulation against them. If the two disagree by more than the tolerance in the right column, **the model is wrong, not the theory** — this geometry is well inside the range where the analytic result is reliable.

quantity	closed form	accept if HFSS gives
Realized gain	13.4 dBi	11.9 – 14.9 dBi
Aperture bound, $4\pi A/\lambda^2$	16.3 dBi	never exceeded — if it is, stop and find the error
HPBW, E-plane ($\phi = 90^\circ$)	34.2°	30° – 38°
HPBW, H-plane ($\phi = 0^\circ$)	36.2°	32° – 40°
First sidelobe	–13 dB (E), lower in H	≤ -12 dB
S_{11} , 2.400–2.484 GHz	set by the probe, not the flare	≤ -10 dB after tuning
TE_{10} cutoff	1.735 GHz	port mode 1 must be propagating
Next mode, TE_{20}	3.470 GHz	must stay cut off — check port modes

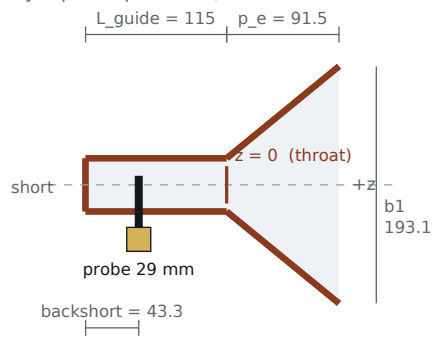
Coordinates used throughout

- **+z** is boresight, out of the mouth. **z = 0 at the throat**, where the flare starts — so the back short is at negative z and the aperture at positive z.
- **x** along the broad wall a (86.4 mm) → this is the **H-plane**, $\phi = 0^\circ$.
- **y** along the narrow wall b (43.2 mm) → E-field direction, the **E-plane**, $\phi = 90^\circ$.

NOTES

- The build rotates all three horns 90° on the frame so b_1 ends up horizontal. That is a mounting choice and changes nothing in this model — simulate it in the natural orientation and rotate the result in your head.

E-plane section (the y-z plane, $\phi = 90^\circ$). Drawn to scale; all dimensions mm.



NOTES

Part I — building the model

1 New design, and get the units right first

- **Project → Insert HFSS Design.**
- **HFSS → Solution Type → Driven Modal.** Modal is the right choice for a single coax or waveguide feed; Driven Terminal is for multi-conductor lines where you want voltages per conductor.
- **Modeler → Units → mm.** Do this before drawing anything. Changing units later rescales nothing and silently reinterprets every number you have typed.

2 Enter the geometry as variables, not as numbers

Type a name such as a1 into any dimension box and HFSS offers to create it. Do this for all of them now: probe length and backshort are the two you will sweep, and a model built from literals cannot be swept at all.

variable	value	what it is
freq	2.45 GHz	design frequency
a	86.40 mm	guide broad wall, WR-340
b	43.20 mm	guide narrow wall
a1	263.79 mm	aperture, H-plane
b1	193.14 mm	aperture, E-plane
pe	91.55 mm	axial flare length ($p_e = p_h$)
Lg	115 mm	guide section length behind the throat
bshort	43.32 mm	probe to back wall = $\lambda_g/4$ — sweep this
plen	29 mm	probe into the guide — sweep this
dpin	1.5875 mm	probe rod, 1/16 inch brass
ddiel	5.318 mm	PTFE outside diameter, see step 5
clear	40 mm	air-box clearance ($\lambda/4 = 30.6$ mm minimum)

3 The guide, the flare, and why you draw the air

Model the air, not the copper

What radiates is the field in the volume inside the horn. Drawing 0.5 mm copper sheet and asking HFSS to mesh it wastes an enormous number of tetrahedra resolving metal that carries no field to speak of. Draw the interior air, then put a perfectly conducting **sheet** where each wall goes. Electrically that is exactly what a soldered copper horn is at 2.45 GHz, and it solves in a fraction of the time.

- **Draw → Box** for the guide: position $(-a/2, -b/2, -Lg)$, size (a, b, Lg) . Name it guide_air.
- **Draw → Rectangle** at the throat, $z = 0$, spanning $a \times b$ centred on the axis. Draw a second rectangle at $z = pe$ spanning $a1 \times b1$.

NOTES

- Select both rectangles, then **Modeler → Surface → Connect**. That lofts a solid between the two profiles — this is the flare. Name it flare_air.
- Select both solids → **Modeler → Boolean → Unite**. Unite keeps the name of the first object selected, so **rename the result horn_air** — the rest of this procedure calls it that. Assign material **vacuum**.

Sanity check the loft before going on: the flare half-angles should come out 39.3° in the E-plane and 44.1° in the H-plane. If Connect refuses, or produces a twisted solid, the two rectangles are not coaxial or not parallel.

4 Turn the walls into perfect-E sheets

- Select horn_air, switch to **face selection**, and pick every outside face **except the aperture** at $z = p_e$. That is: four flare panels, four guide walls, and the back short.
- **Modeler → Surface → Create Object From Face**. You now have sheet objects lying exactly on those walls.
- **Delete horn_air**. It was scaffolding. The air box in step 7 will fill that volume, and the sheets stay behind as the metal.
- Select all the new sheets → **right-click → Assign Boundary → Perfect E**. Name it walls.

Second pass, later: to see conductor loss, assign **Finite Conductivity** with copper (5.8×10^7 S/m) instead of Perfect E. Expect it to cost a few hundredths of a dB at this frequency — worth knowing, not worth waiting for on the first solve.

5 The coax feed — the part that decides your S11

The real probe is a 1/16 inch brass rod, 1.587 mm across. If you model the rod at that diameter you must size the PTFE around it to keep the stub at 50 Ω , or you have built an impedance step into your own model and will spend an evening tuning it out:

$$Z_0 = (138/\sqrt{\epsilon_r}) \cdot \log_{10}(D/d) = 50 \, \Omega \text{ with } \epsilon_r = 2.1 \rightarrow D/d = 3.350 \\ \rightarrow D = 5.318 \text{ mm}$$

- **Pin**. Cylinder, radius $d_{\text{pin}}/2$, axis along **+y**, from $y = -b/2 - 10$ (a 10 mm stub outside the wall, where the port goes) up to $y = -b/2 + p_{\text{len}}$. Centre it at $x = 0$, $z = -L_g + b_{\text{short}}$. Material: **pec**. Name pin.
- **Dielectric**. Cylinder, radius $d_{\text{diel}}/2$, same axis, from $y = -b/2 - 10$ to $y = -b/2$ (it stops at the wall). Material **Teflon (ϵ_r 2.1)**. Name coax.
- **Boolean → Subtract**: coax minus pin, with **“clone tool objects” ticked** so the pin survives.
- Assign **Perfect E** to the outer curved face of coax — that is the shield.
- The wall sheet the probe passes through needs a clearance hole: subtract coax from that sheet, or draw the sheet with the hole already in it. A probe shorted to the wall gives a beautiful and completely fictional S_{11} .

Geometry check: the probe tip sits at $y = 7.4$ mm and the far wall is at $y = 21.6$ mm, so there is 14.2 mm of clearance. It must stay positive at every value of p_{len} you sweep.

6 Wave port on the end of the stub

NOTES

- Select the **flat annular end face** of coax at $y = -b/2 - 10$.
- **Assign Excitation → Wave Port**. Modes: **1**. Add an **integration line** from the pin surface radially out to the shield — this fixes the sign and the impedance reference.
- Leave **Renormalize to 50 Ω** on.

The port must enclose the coax and nothing else

A wave port on a coax should be exactly the dielectric annulus. If you accidentally draw it larger — onto the air box, say — it stops being a TEM coax port and starts supporting hollow-waveguide modes, and your S_{11} becomes meaningless. After the first solve, open **HFSS → Results → Solution Data → Port Field Display** and look at mode 1: it must be the radial TEM pattern of a coax.

7 Air box and radiation boundary

Draw → Box enclosing everything, material **vacuum**. With clear = 40 mm that is roughly:

axis	from	to	why
x	-172 mm	172 mm	aperture half-width + clearance
y	-137 mm	137 mm	includes room for the coax stub below
z	-155 mm	132 mm	behind the short, and in front of the mouth

- **Subtract** coax and pin from the air box, cloning the tools, so no two solids overlap.
- Select all six outer faces → **Assign Boundary → Radiation**.

$\lambda/4$ at 2.45 GHz is 30.6 mm, and that is the minimum standoff for a radiation boundary — closer and it reflects, which moves both the pattern and S_{11} . 40 mm gives some margin. If the model is too big to solve, the right fix is a **PML** (which works down to about $\lambda/8$), not a smaller radiation box.

NOTES

Part II — solving and checking

8 Solution setup and sweep

setting	value	why
Solution Frequency	2.45 GHz	the adaptive mesh is built at this one frequency
Maximum Number of Passes	20	a ceiling, not a target — it should converge before this
Maximum Delta S	0.02	0.02 is the usual working value; tighten to 0.01 for the final run
Sweep type	Interpolating	far fewer solves than Discrete for a smooth S-curve
Sweep range	2.20 – 2.70 GHz	wide enough to see the resonance move while tuning
Points	501	interpolating decides its own solve points; this is the output grid

Sweeping well below the band matters while you tune: a probe that is mistuned shows up as a dip at 2.2 or 2.7 GHz, and if you only plot 2.4–2.484 you see a flat, featureless –3 dB and no clue which way to move.

9 Far-field setup

- **Radiation → Insert Far Field Setup → Infinite Sphere.**
- **3D:** θ 0→180 step 2°, ϕ 0→360 step 2°. Name it ff_3d.
- **H-plane cut:** $\phi = 0$ fixed, θ –180→180 step 1°. Name it cut_H.
- **E-plane cut:** $\phi = 90$ fixed, θ –180→180 step 1°. Name it cut_E.

Getting the two planes the wrong way round is the most common reporting error in horn work, and it is easy to catch: **the E-plane is the narrower beam** here (34.2° against 36.2°), because the H-plane carries the cosine taper of the TE_{10} mode. If your $\phi = 0$ cut is the narrow one, your guide is rotated 90° from the convention on the previous page.

10 Analyse, then check the solver before you believe it

HFSS → Analyze All. Then, in this order:

#	check	where	what you want
1	Convergence	Results → Solution Data → Convergence	ΔS below target, and the last two or three passes changing very little — one pass sneaking under the threshold is luck, not convergence
2	Port mode	Solution Data → Port Field Display	Mode 1 is coax TEM. If it looks like a waveguide mode, the port is wrong

NOTES

#	check	where	what you want
3	Mesh count	Solution Data → Profile	Tens to a couple of hundred thousand tetrahedra. Millions means something is over-refined — usually a thin solid
4	Gain vs the bound	Far field → Gain Total, dB	Must be under 16.3 dBi . Above it, the model is wrong
5	Then the numbers	the table on page 1	Gain, both beamwidths, sidelobes, S_{11}

1 1 Tune the two knobs

The flare sets the pattern and the gain, and it is already optimal — do not touch it. The **match** is set entirely by the feed, and there are exactly two variables:

- **plen** (probe depth) — sets coupling, and mostly moves the real part of Z_{in} .
- **bshort** (probe to back wall) — mostly trims the reactance.

Optimetrics → Add → Parametric. Sweep plen from 24 to 34 mm in 1 mm steps and bshort from 37 to 49 mm in 2 mm steps, and plot a family of S_{11} curves. You are looking for the pair that puts the dip in the middle of 2400–2483.5 MHz and keeps it below –10 dB across the whole span. Two variables and about sixty solves is an overnight job, and it is the only part of this that genuinely needs the solver.

What the simulation is really buying you

The gain and the pattern you could have had from horn.py in a millisecond, and the agreement between them is a check on the model, not a discovery. The probe match is the part with no useful closed form — and it is also the one thing the build notes say you can tune on the bench against detection SNR without a VNA. Simulating it first means you arrive at the bench already close.

1 2 The second model: TX-to-RX isolation

A separate project, and the one antenna question this radar has that a single horn cannot answer. The build mounts TX 290 mm above the RX row and needs $S_{21} \leq -35$ dB; the leakage that gets through is ~46 dB stronger than the drone echo and sets the whole dynamic-range problem downstream.

- Duplicate the horn (**Edit → Duplicate → Along Line**), offset **290 mm along y** — the E-plane direction, which is how the frame stacks them.
- Two wave ports, one per horn. Grow the air box so it still clears $\lambda/4$ from **both** apertures.
- Plot S_{21} across the band. Expect a large model and a long solve — this is the one worth running overnight.

NOTES

- ## Record what you get

quantity	closed form	HFSS	measured
Realized gain, dBi	13.4		
HPBW E-plane, °	34.2		
HPBW H-plane, °	36.2		
First sidelobe, dB	−13		
S_{11} at 2.400 GHz	−		
S_{11} at 2.442 GHz	−		
S_{11} at 2.484 GHz	−		
Best plen / bshort, mm	29 / 43.3		
TX-RX S_{21} at 290 mm, dB	target ≤ -35		